

3.1.2 Amount of substance

When chemists measure out an amount of a substance, they use an amount in moles. The mole is a useful quantity because one mole of a substance always contains the same number of entities of the substance. An amount in moles can be measured out by mass in grams, by volume in dm^3 of a solution of known concentration and by volume in dm^3 of a gas.

3.1.2.1 Relative atomic mass and relative molecular mass

Content	Opportunities for skills development
Relative atomic mass and relative molecular mass in terms of ^{12}C. The term relative formula mass will be used for ionic compounds. Students should be able to: - define relative atomic mass (A_r) - define relative molecular mass (M_r).	

3.1.2.2 The mole and the Avogadro constant

Content	Opportunities for skills development
The Avogadro constant as the number of particles in a mole. The mole as applied to electrons, atoms, molecules, ions, formulas and equations. The concentration of a substance in solution, measured in mol dm^{-3}. Students should be able to carry out calculations: - using the Avogadro constant - using mass of substance, M_r, and amount in moles - using concentration, volume and amount of substance in a solution. Students will not be expected to recall the value of the Avogadro constant.	MS 0.1 Students carry out calculations using numbers in standard and ordinary form, eg using the Avogadro constant. MS 0.4 Students carry out calculations using the Avogadro constant. MS 1.1 Students report calculations to an appropriate number of significant figures, given raw data quoted to varying numbers of significant figures. Students understand that calculated results can only be reported to the limits of the least accurate measurement.

3.1.2.3 The ideal gas equation

Content	Opportunities for skills development
The ideal gas equation $pV = nRT$ with the variables in SI units. Students should be able to: • use the equation in calculations. Students will not be expected to recall the value of the gas constant, R.	AT a, b and k PS 3.2 Students could be asked to find the M_r of a volatile liquid. MS 0.0 Students understand that the correct units need to be in $pV = nRT$. MS 2.2, 2.3 and 2.4 Students carry out calculations with the ideal gas equation, including rearranging the ideal gas equation to find unknown quantities.

3.1.2.4 Empirical and molecular formula

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Empirical formula is the simplest whole number ratio of atoms of each element in a compound. Molecular formula is the actual number of atoms of each element in a compound. The relationship between empirical formula and molecular formula. Students should be able to: • calculate empirical formula from data giving composition by mass or percentage by mass • calculate molecular formula from the empirical formula and relative molecular mass.	AT a and k PS 2.3 and 3.3 Students could be asked to find the empirical formula of a metal oxide.

3.1.2.5 Balanced equations and associated calculations

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Equations (full and ionic). Percentage atom economy is: $\frac{\text{molecular mass of desired product}}{\text{sum of molecular masses of all reactants}} \times 100$ Economic, ethical and environmental advantages for society and for industry of developing chemical processes with a high atom economy. Students should be able to: • write balanced equations for reactions studied • balance equations for unfamiliar reactions when reactants and products are specified. Students should be able to use balanced equations to calculate: • masses • volumes of gases • percentage yields • percentage atom economies • concentrations and volumes for reactions in solutions.	AT a, d, e, f and k PS 4.1 Students could be asked to find: • the concentration of ethanoic acid in vinegar • the mass of calcium carbonate in an indigestion tablet • the M_r of MHCO_3 • the M_r of succinic acid • the mass of aspirin in an aspirin tablet • the yield for the conversion of magnesium to magnesium oxide • the M_r of a hydrated salt (eg magnesium sulfate) by heating to constant mass. AT a and k Students could be asked to find the percentage conversion of a Group 2 carbonate to its oxide by heat. AT d, e, f and k Students could be asked to determine the number of moles of water of crystallisation in a hydrated salt by titration. MS 0.2 Students construct and/or balance equations using ratios. Students calculate percentage yields and atom economies of reactions. MS 1.2 and 1.3 Students select appropriate titration data (ie identify outliers) in order to calculate mean titres. Students determine uncertainty when two burette readings are used to calculate a titre value.

Content	Opportunities for skills development
Required practical 1 Make up a volumetric solution and carry out a simple acid–base titration.	